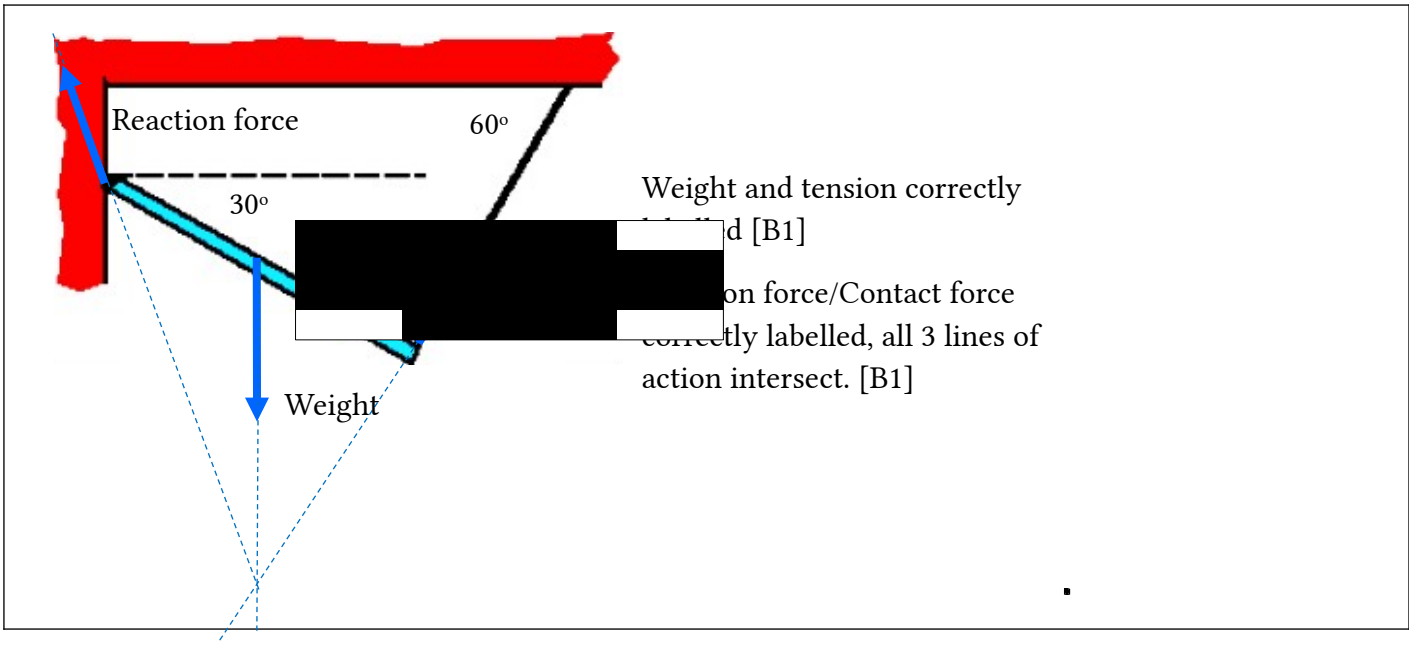
2026 H2 Physics Prelim Paper 3 Solutions

1 (a) The moment of a force is defined as the <u>product of the force</u> and the <u>perpendicular distance from the pivot</u> to the <u>line of action of the force</u>. B1

(b)(i)



Weight and tension correctly labelled [B1]

Reaction force/Contact force correctly labelled, all 3 lines of action intersect. [B1]

(ii) Take moments about the hinge,

$30 \times 9.81 \times \cos 30^o (\frac{L}{2}) = T \times L$ M1

$T = 127.4 \approx 127\ N$ A0

(iii) $R_X = T\cos 60^o = 127.4\cos 60^{oo} = 63.7N$ M1

$R_Y + T_Y = W$

$R_Y + 127\sin 60^\circ = 30(9.81)$

$R_Y = 184\ \text{N}$ M1

$R = \sqrt{R_X^2 + R_Y^2} = \sqrt{63.7^2 + 184^2} = 194.7 \approx 195\ N$ A1

$\tan\theta = \frac{184}{63.7} \Rightarrow \theta = 70.9^o$ A1

above the horizontal

2 (a) The total momentum of a system remains <u>constant</u> if no <u>resultant external force</u> acts on the system. B1

(b) (i) Relative speed of approach = 6 – 0 = 6 ms^{-1} C1

Relative speed of separation = 4 – (-2) = 6 ms^{-1}

Since they are equal, the collsion is elastic A1

(ii) Using conservation of momentum:

$$m_x u_x + m_y u_y = m_x v_x + m_y v_y$$

$$3.0(6.0) + 0 = 3.0(-2.0) + m(4.0)$$ C1

A0

$\Rightarrow$ m = 6.0 kg

OR

Using conservation of kinetic energy:

$$KE_{initial} = KE_{final}$$

$$\frac{1}{2}(3.0)(6.0)^2 + 0 = \frac{1}{2}(3.0)(2.0)^2 + \frac{1}{2}(m)(4.0)^2$$ C1

A0

$$m = 6.0 \text{ kg}$$

(iii)1

Cart Y:

$$\langle F \rangle = \frac{6.0(4.0) - 0}{40 \times 10^{-3}}$$

$$= 600 \text{ N}$$

OR

Cart X:

$$\langle F \rangle = \frac{3.0(-2.0) - 3.0(6.0)}{40 \times 10^{-3}}$$

$$= -600 \text{ N}$$

C1

A1

(iii)2

$$\langle P \rangle = \frac{\text{Gain in KE}}{t}$$

$$= \frac{\frac{1}{2}(6.0)(4.0)^2 - 0}{40 \times 10^{-3}}$$

$$= 1200 \text{ W}$$

W

C1

A1

OR

$$\langle P \rangle = \langle Fv \rangle = \langle F \rangle \left(\frac{u+v}{2} \right) = 600 \left(\frac{0+4.0}{2} \right) = 1200 \text{ W}$$

(iii)3

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2}(3.0)(6.0)^2 + 0 = \frac{1}{2}(3.0)(2.0)^2 + \frac{1}{2}(6.0)(2.0)^2 + EPE$$

$$EPE = 36 \text{ J}$$

C1

A1

3 (a) Gravitational force provides for centripetal force.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$ M1

$$\frac{GM}{r} = v^2$$

$$v = \sqrt{\frac{GM}{r}}$$ A0

(b)(i) $$v = \frac{2\pi r}{T} \Rightarrow T = \frac{2\pi r}{v}$$ C1

$$\frac{T_A}{T_B} = \frac{r_A}{r_B} \times \frac{v_B}{v_A}$$

$$= \frac{r_A}{r_B} \times \sqrt{\frac{r_A}{r_B}} \qquad \text{since } v = \sqrt{\frac{GM}{r}} \propto \sqrt{\frac{1}{r}}$$ C1

$$= \frac{1.3 \times 10^8}{2.2 \times 10^{10}} \times \sqrt{\frac{1.3 \times 10^8}{2.2 \times 10^{10}}}$$

$$= 4.5 \times 10^{-4}$$ A1

(ii)

$$\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega}$$

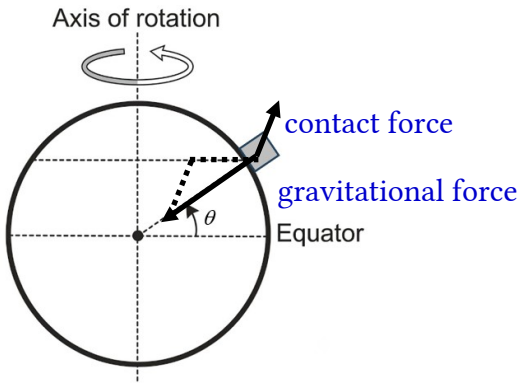
$$T_A = \frac{2\pi}{1.7 \times 10^{-4}} = 3.7 \times 10^{-4}$$ C1

$$\frac{T_A}{T_B} = 4.5 \times 10^{-4}$$

$$T_B = \frac{T_A}{4.5 \times 10^{-4}}$$

$$= \frac{3.7 \times 10^{-4}}{4.5 \times 10^{-4}}$$

$$= 8.2 \times 10^7 \text{ s}$$ A1



• Gravitational force pointing towards the centre of planet B1

• Contact force by ground on block. (Accept if normal contact force and friction force are drawn separately). B1

Resultant of the 2 forces must act towards centre of circle

4 (a)

$$\omega^2 = \frac{k}{m} \Rightarrow \left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$$

$$T^2 = \frac{4\pi^2 m}{k} \Rightarrow T \propto \sqrt{\frac{m}{k}}$$ C1

$$\frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1} \times \frac{k_1}{k_2}} = \sqrt{\frac{2m}{m} \times \frac{k}{0.5k}} = \sqrt{4} = 2$$ A1

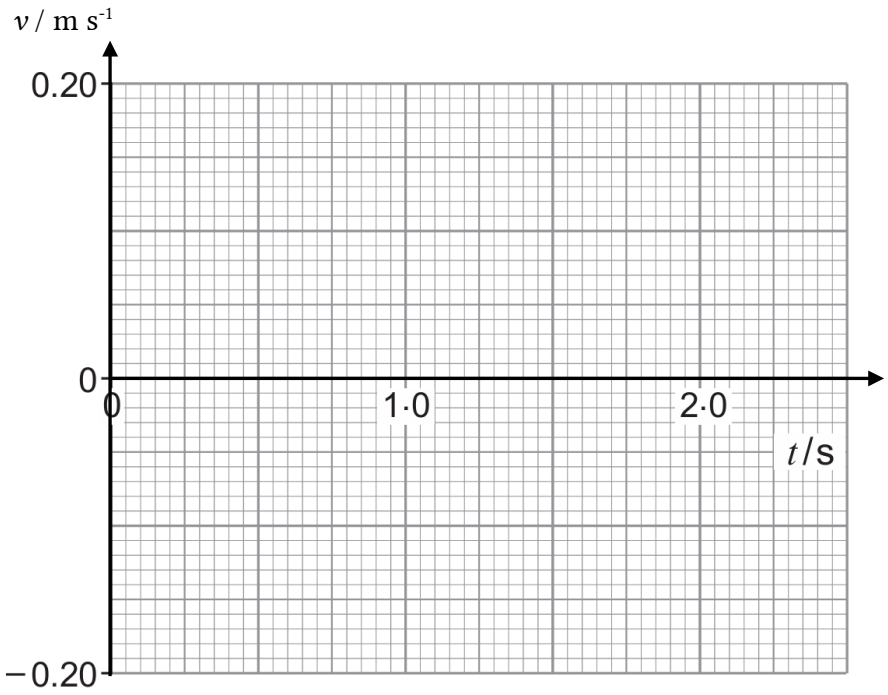
(b)(i)

$$v_{max} = \omega x_o = \left(\frac{2\pi}{T}\right) x_o$$ M1

$$= \left(\frac{2\pi}{1.20}\right)\left(30.0 \times 10^{-3}\right)$$

$$= 0.157 = 0.16 \text{ m s}^{-1}$$ A0

(b)(ii)



inverted sine graph B1
period 1.2 s, 2 oscillations drawn, peak value 0.16 m s^{-1} B1

(b)(iii)

$$v = -0.157 \sin \omega t = -0.157 \sin\left(\frac{2\pi}{1.20}\right)(3.50)$$ C1

$$= 0.079 \text{ m s}^{-1}$$ A1

(Alternative: Speed the same as at 1.1 s and 2.3 s. Speed = 0.079 m s^{-1})

(b)(iv) T = 1/1.20 = 0.833 Hz, 1.67 Hz is twice the oscillation frequency. B1
This is correct because KE = ½ mv^2 and v^2 will have two [equal] positive peaks per oscillation cycle. B1

5 (a)(i) When one polaroid is rotated, resultant intensity of bright fringes decreases but the resultant intensity of dark fringes increases. The pattern becomes less defined as the contrast between the maxima and minima decreases. B1
When the axes of polaroids are perpendicular to each other, the fringe pattern disappears to give uniform illumination. B1

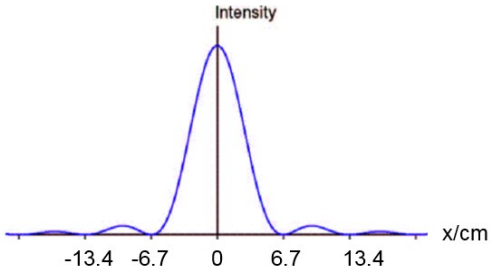
(a)(ii) 1.

$$\sin\theta = \frac{\lambda}{b} = \frac{644 \times 10^{-9}}{2.4 \times 10^{-5}}$$ C1

$$\Rightarrow \theta = 1.54^{o}$$

tan1.54° = x /2.5 => x = 0.067 m C1
Width = 2x = 2(0.067) = 0.134 m A1

(a)(ii) 2.



- At least 3 maxima, symmetrical about O
- Correct shape, with decreasing intensity
- Correct spacing
- Correct label (Any missing above - minus 1 mark)

B3

(b)(i) According to Rayleigh criterion, when the central maximum of one image falls on the first minimum of another image, the images are said to be just resolved. B1

(b)(ii) Since 2.0° is greater than the minimum angle of 1.54° for the 2 images to be revolved, patterns resolved. B1

6a	$Q = CV = (220 \times 10^{-6})(6.0) = 1.32 \times 10^{-3}$ C	C1 A1
b	$E = \frac{1}{2} C V^2 ¿ \frac{1}{2}(220 \times 10^{-6})(6.0)^2 = 3.96 \times 10^{-3}$ J $= 4.0 \times 10^{-3}$ J	C1 A1
ci	Tangent drawn correctly	B1
ii	$V=V_0(1-e^{-t/RC})$ so the initial gradient is $\left.\frac{dV}{dt}\right\|_{t=0} = \frac{V_0}{RC}$.tangent = 1V s^{-1} hence $CR = 6.0$ s (accept 5.4 to 6.6s)	C1 A1
iii	$CR = 6.0R = \frac{6.0}{C} ¿ \frac{6.0}{220 \times 10^{-6}} = 2.7 \times 10^4 \Omega = 27$ kΩ	C1 A1

7 (a) The electromotive force (e.m.f.) of a source is defined as the amount of energy transferred from non-electrical forms of energy to electrical energy per unit charge as the charge passes through a complete circuit. [B1]

The potential difference *V* between two points of a circuit is defined as the amount of electrical energy converted to **other** forms of energy **per unit charge** moved between these two points. [B1]

(i)

Total resistance, $R_T = 1.0 + \dfrac{1}{\frac{1}{2.0} + \frac{1}{6.0}} = 2.5\ \Omega$ [M1]

$E = I R_T$ [M1]

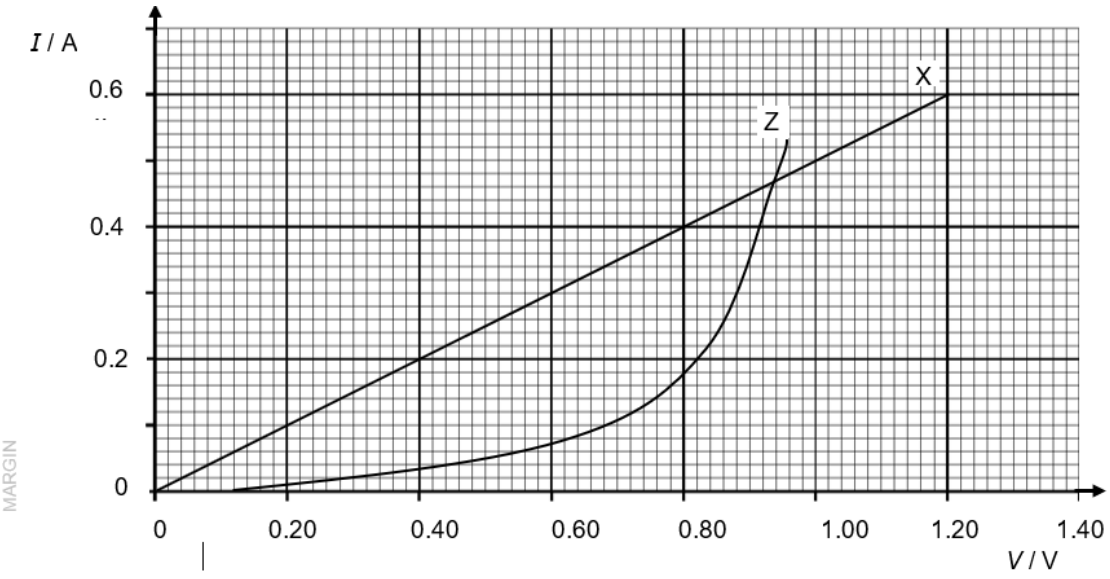
I = 1.5 / 2.5 = 0.60 A (Shown) [A0]

(ii) Energy dissipated = $I^2 r t$ [M1]

$= (0.60)^2 (1.0)(8.0 \times 60) = 172.8 = 170$ J [B1]

(iii) When temperature increased and resistance of Z decreases, the total current in the cell will increase. Thus, the power dissipated in the cell increases. [B1]

(iv)

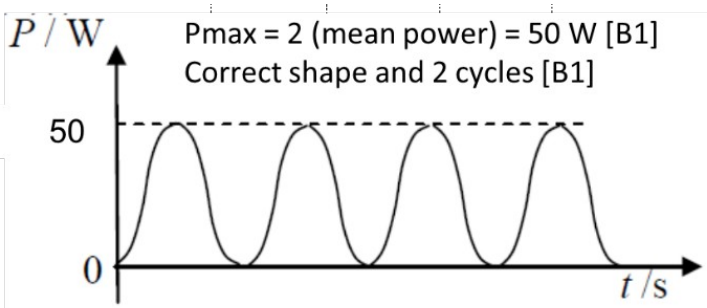


Since the components are connected in series, the current passing through the components are the same, draw a horizontal line to find the total p.d. across equal to 1.5 V. [B1]

p.d across the X and 1.0 Ω is 0.66V, and p.d across Z is 0.84V [B1]

v Thermistor [B1]

7 **(c)i**



Pmax = 2 (mean power) = 50 W [B1]
Correct shape and 2 cycles [B1]

cii Sketch of P-t graph with llustration of area of rectangle of height $1/2P_o$.
B1
Explain that the area of the rectangle is the same as the area of the P-t curve
B1

(d) (i) $\frac{Ns}{Np} = \frac{Vs}{Vp}$

Vs = 1/45 (240) = 5.33V

Hence $V_o = \sqrt{2} \times 5.33 = 7.54$ V [B1]

Assumption : Transformer is ideal / 100% efficiency [B1]

(ii) The voltage step down to the secondary coil is sinusoidal but with the diode in place, the voltage across the battery is half wave rectified . This imply that the time that the battery get charged is only half of the time voltage is step down across the secondary coil . Hence the energy received by the battery is reduced by half. [B1]

(iii)

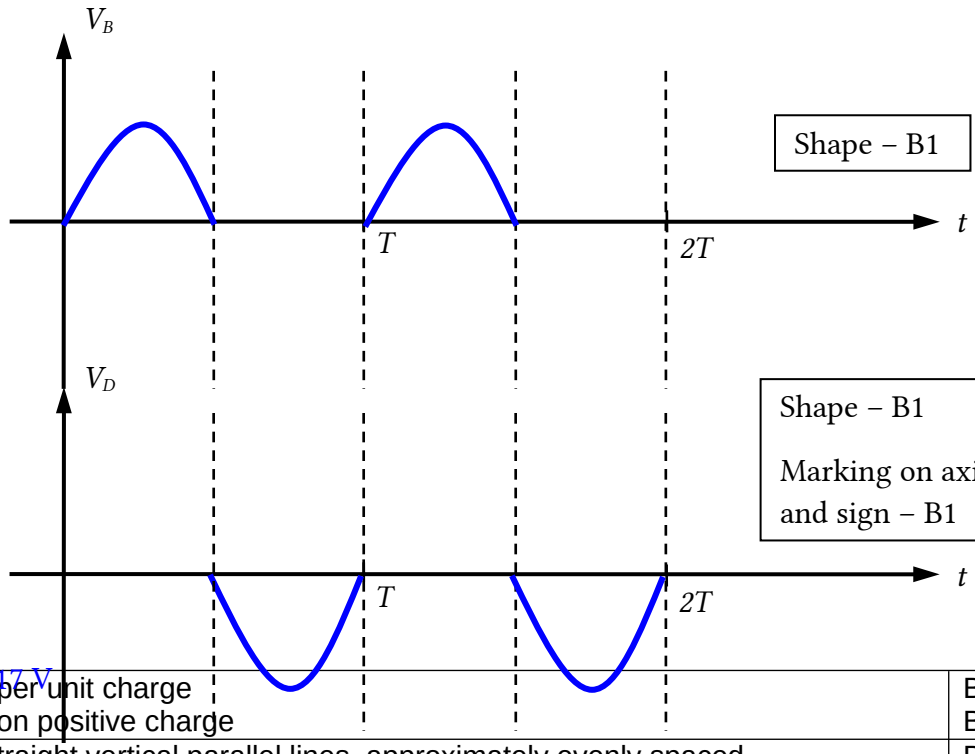
Without diode , $P = (5.3)^2/R$

With diode , power delivered = $P/2 = V'^2_{rms}/R$, $28.1/2R = V'^2_{rms}/R$ [B1]

Hence V'_{rms} $\sqrt{(28.1/2)} = 3.74$ V

(iv)

(v)



1]

8a	force per unit charge force on positive charge	B1 B1
bi	four straight vertical parallel lines, approximately evenly spaced arrows downwards	B1 B1
ii	$E = \frac{V}{d} E = \frac{430}{0.067} = 6.4 \times 10^3 \text{ N C}^{-1}$	C1 A1
iii	smooth curve within plates and straight lines outside plates direction of deflection shown as upwards	B1 B1
ci	into the page	B1
ii	forces are in opposite directions and are equal	B1
Iii	$Eq = Bqv \quad B = \frac{E}{v} = \frac{6.4 \times 10^3}{2.6 \times 10^7} = 2.5 \times 10^{-4} \text{ T}$	C1 A1
d	induced e.m.f. proportional to rate of change of magnetic flux linkage	B1 B1
e	Flux linkage $= BAN = \pi \times 10^{-3} \times 2.8 \times \pi \times \left(1.6 \times 10^{-2}\right)^2 \times 85 = 6.0 \times 10^{-4} \text{ Wb}$	B1
f	$\text{e.m.f.} = \frac{\Delta N\phi}{\Delta t} \; ¿ \; \frac{\left(6.0 \times 10^{-4} \times 2\right)}{0.30}$	C1

	$= 4.0\ \text{mV}$	A1
g	Sketch: $E = 0$ for $t = 0 \rightarrow 0.3\ \text{s}, 0.6\ \text{s} \rightarrow 1.0\ \text{s}, 1.6\ \text{s} \rightarrow 2.0\ \text{s}$ $E = 4\ \text{mV}$ for $t = 0.3\ \text{s} \rightarrow 0.6\ \text{s}$ (either polarity) $E = 2\ \text{mV}$ for $t = 1.0\ \text{s} \rightarrow 1.6\ \text{s}$ with opposite polarity	-1 per error